

Pratyay Roy

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Summary

Full Stack Software Engineer specializing in robust backend architectures (**Laravel**, **FastAPI**) and modern frontends (**React**). Experienced in architecting end-to-end applications, designing transactional **REST APIs**, and building scalable data layers with **MySQL**. **Microsoft Azure Certified (AZ-900)** with practical experience in cloud deployment.

Education

- **Techno India University** **June 2022 – May 2026**
- B.Tech in CSE (AI/ML) **Kolkata, West Bengal**
- **CGPA: 7.6 / 10**
- **Relevant Coursework:** Operating Systems (OS) • Database Management Systems (DBMS) • Computer Networks (CN)
 - Data Structures & Algorithms (DSA)

Technical Skills

- **Languages:** Java (Core Java, Java 21), PHP, Python, SQL
- **Frontend:** HTML5, CSS3, Bootstrap, JavaScript (ES6+), TypeScript, React, AJAX, JQuery
- **Frameworks and Libraries:** Laravel 12, Inertia.js, FastAPI, NumPy, Pandas, OpenCV, MediaPipe, EasyOCR, PyAutoGUI, Matplotlib
- **Backend and Databases:** MySQL, PostgreSQL, HTTP Middleware, Eloquent ORM, Sanctum, Pydantic
- **Cloud, DevOps and Tools:** Microsoft Azure (AZ-900 Azure Fundamentals), Git, GitHub, Postman, Docker
- **Core Concepts:** Object-Oriented Programming (OOP), Database Management Systems (DBMS), Operating Systems, Data Structures and Algorithms (DSA)

Experience

- **Frontend Developer** — InAmigos Foundation (Contract) **May 1 – May 15, 2026**
 - Executed a rapid 15-day development sprint, completing 5 technical delivery milestones including codebase auditing, frontend refactoring, and automated production deployment.
 - Architected and deployed a multi-page foundational web platform using **React** and **Netlify**, implementing modular component structures, state management, and optimized build pipelines.
 - Conducted comprehensive UI/UX audits on existing web systems to identify architectural bugs, accessibility issues, and responsive layout flaws, compiling structural hotfixes for cross-platform stability.

Projects

- **Nutri Scan: Health Data Processing Platform** — Python, FastAPI, PostgreSQL, EasyOCR **October 2025**
 - Designed a normalized SQL relational database schema with **Pydantic**-driven validation and optimized indexing, reducing health trend data retrieval time by **25%** across **50+ medical test formats**.
 - Built a **React** frontend integrating **FastAPI** auto-generated **OpenAPI** endpoints to deliver real-time health trend visualizations and streamline API responses.
- **MediTrack: Hospital & Clinic Management System** — PHP, Laravel 13, Sanctum, MySQL, Blade **June 2026**
 - Architected a role-based Hospital Management System in **Laravel 13** with **Sanctum**, building **3 role-specific dashboards** across **6 Eloquent models** and a full **REST API** with end-to-end **CRUD** coverage over **5+ resource endpoints**.
 - Integrated the **openFDA Drug Label API** into the prescription workflow enabling real-time lookup across **100K+ indexed drug records**; automated **Mailable** notifications on appointment confirmation and prescription dispatch, reducing manual communication overhead by **100%**.
 - Enforced **3-layer role middleware** with **2 custom server-side validation rules** and a **bilingual (EN/BN) i18n** system; built a patient medical report upload pipeline supporting **3 file formats** (PDF/JPG/PNG) with persistent **MySQL** storage.

Certifications

- **Microsoft Azure Fundamentals (AZ-900)** – Microsoft
- **Cybersecurity** – Campus Technology, Times of India Group
- **TCS iON Career Edge** – Tata Consultancy Services